
Chapter 2: Power Play (Exponents and Powers)

Worksheet 2: Laws of Exponents

Learning Objective: Students will apply laws of exponents (product, quotient, power of a power, zero exponent, negative exponents); express numbers in standard form; and apply exponents to very large and very small quantities.

Worked Example

Simplify: $\frac{3^5 \times 3^{-3}}{3^2}$ and express in standard form: 0.000045

Solution:

$$\frac{3^5 \times 3^{-3}}{3^2} = \frac{3^{5-3}}{3^2} = \frac{3^2}{3^2} = 3^0 = 1$$

Standard form: $0.000045 = 4.5 \times 10^{-5}$

Questions 1–5: Easy / Recall

Q1. Simplify: $2^3 \times 2^4$

Q2. Simplify: $\frac{5^7}{5^4}$

Q3. Find the value of $7^0 + 3^0 - 1^0$.

Q4. Write 2^{-3} as a fraction.

Q5. Simplify: $(4^2)^3$

Questions 6–10: Basic Application

Q6. Simplify: $\left(\frac{2}{3}\right)^{-2}$

Q7. Express in standard form: (a) 6,750,000 (b) 0.00000032

Q8. Simplify: $\frac{(-2)^4 \times 3^3}{6^2}$

Q9. If $2^x = 128$, find x .

Q10. Simplify: $\left(\frac{5^3 \times 5^{-2}}{5}\right)^2$

Questions 11–15: Practice and Skill Building

Q11. Simplify and express as a power of a single base: $\frac{2^8 \times 3^4 \times 6^2}{4^3 \times 9^2}$

Q12. The mass of the Earth is approximately 6×10^{24} kg and the mass of the Moon is 7.4×10^{22} kg. How many times heavier is the Earth than the Moon?

Q13. Simplify: $(3^{-1} + 4^{-1})^{-1}$

Q14. Find x : $3^{2x-1} = 27^{x-1}$

Q15. Evaluate: $\frac{(0.01)^3}{(0.1)^4}$

Questions 16–18: Higher Order Thinking

Q16. A bacterium doubles every hour. Starting with 1 bacterium, how many are there after 20 hours? Express in standard form.

Q17. Simplify: $\frac{(x^2y^{-1})^3 \times (x^{-1}y^2)^2}{x^4y}$

Q18. If $a = 2^p$ and $b = 2^q$, express $\frac{a^3}{b^2} \times \frac{b^4}{a}$ as a power of 2 in terms of p and q .

Questions 19–20: Real-Life Applications

Q19. The diameter of a human hair is about 7×10^{-5} m. The diameter of a hydrogen atom is 1.06×10^{-10} m. How many hydrogen atoms placed side by side equal the diameter of one hair?

- Q20.** India's national budget in 2024 was approximately Rs. 4.72×10^{13} (in paise). Express this in rupees in standard form. If the population is 1.4×10^9 , find the per-person share in rupees.

Reflection Question: Why is $a^0 = 1$ for any non-zero a ? Explain using the quotient rule of exponents.